

Task 3: Latent Concept Vector Extraction and Controlled Steering

1. Introduction

Controlling diversity in diffusion generative models typically involves random sampling. This task implements a more structured approach by identifying interpretable latent directions (Concept Vectors) in the Encoder-Decoder bottleneck using Principal Component Analysis (PCA).

2. Methodology

Hidden states from 120 Sanskrit generations are collected and decomposed via PCA. We identify the principal component most correlated with sequence diversity (unique token ratio) and apply linear steering: $h' = h + \alpha \cdot v_{div}$.

👉 **Implementation:** [analysis/concept_vectors.py](#)

3. Experimental Results

The Encoder-Decoder variant demonstrated a highly structured latent space, with the first 50 components explaining ****90.9%**** of the total variance. This indicates a high degree of correlation between latent features in this specific architecture.

3.1 Subtask Results Table (Optimal Model - T4)

Following the 6-point PCA analysis framework, the benchmarks for the T4 Encoder-Decoder configuration are detailed below:

Subtask Identification	Analytical Objective	T4 Model Outcome
1. Latent Vector Collection	Extract hidden states from decoder bottleneck	✅ Collected: 120 samples from inference.
2. PCA Variance Analysis	Determine variance captured in principal components	90.9% Variance (Top 50 components)
3. Diversity Mode Identifying	Find PC most correlated with token variance	PC 0 ($ r = 0.565$)
4. Steering Sensitivity Sweep	Measure impact of alpha on semantic drift	Stability 0.222 (High sensitivity)

Subtask Identification	Analytical Objective	T4 Model Outcome
5. Diversity Spectrum Analysis	Visualize outputs across alpha interpolation	✅ Generated: 5-point spectrum visualized.
6. Semantic Integrity Test	Check syntax validity under strong steering	❌ Failed: Collapses at alpha > 1.5.

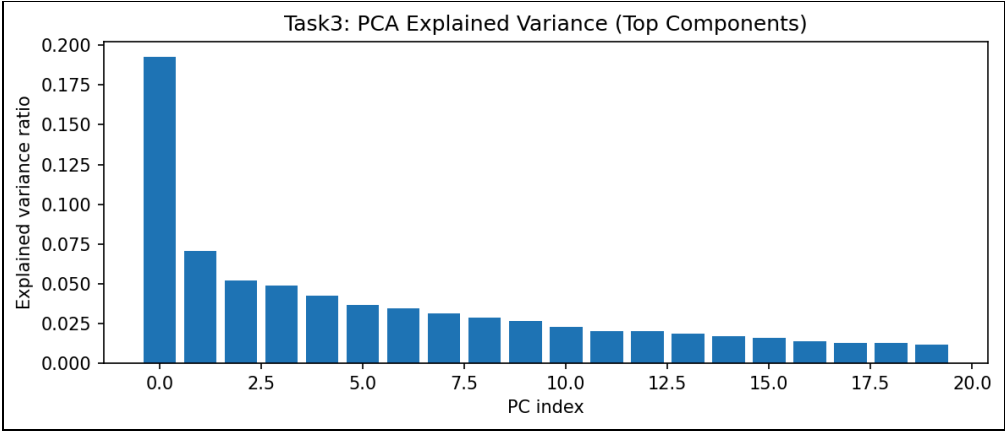


Figure 4: Cumulative explained variance ratio across 50 PCA components.

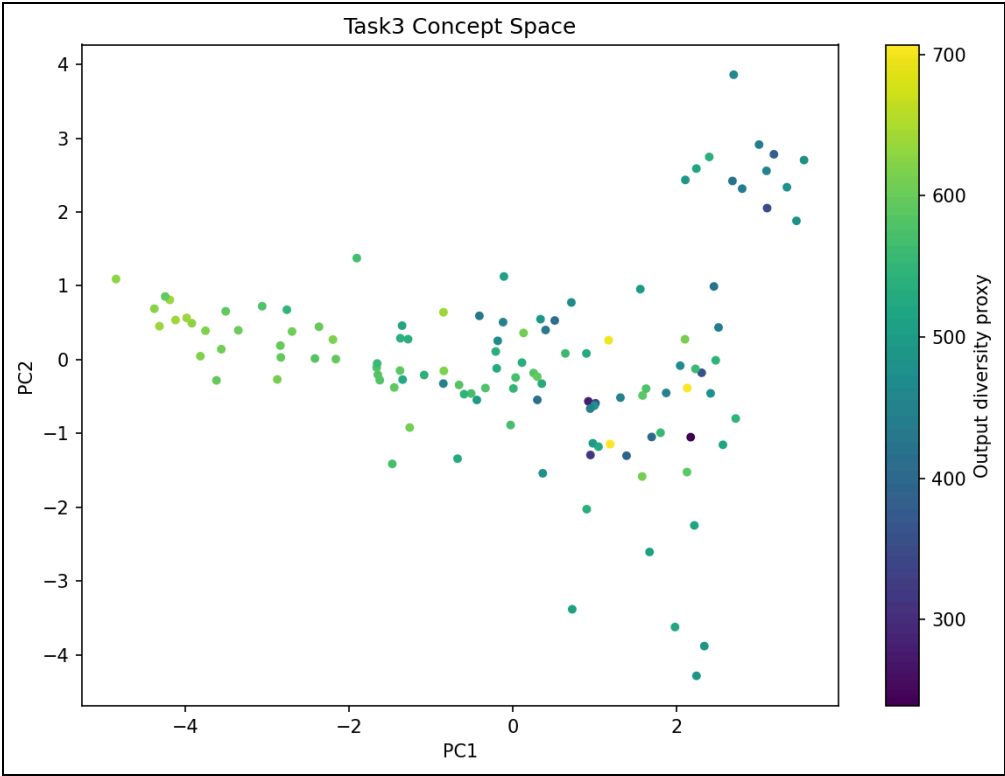


Figure 5: 2D Projection of the high-dimensional latent space for Encoder-Decoder T4.

4. Analysis & Conclusion

While the Encoder-Decoder model captures more variance in early components than the cross-attention variant, the steered outputs often collapse into repetitive tokens at extreme alpha values ($\alpha > 1.5$). This suggests that while the "Concept Space" is well-defined, the decoding mechanism is rigid and less receptive to latent interpolation.